1. Clinical Question & Core Hypothesis

Clinical Question

Pretreatment peritumoral immune patterns (captured by PLSP/TIL measures) should predict cisplatin NAC sensitivity in NMIBC, but some TIL-rich tumors fail to respond—why are morphologically similar immune-infiltrated tumors discordant for chemotherapy response? 2) Tier 1 paths used: (a) immune competence vs exhaustion (PDCD1 as an anchor for T-cell functional state) — because exhausted T cells may be non■functional despite high density, and (b) stromal extracellular matrix barrier (COL1A1 as an anchor for collagen-rich stroma) — because dense collagen can impede drug delivery and immune–tumor contact. These two mechanistic axes directly modulate whether peritumoral lymphocytes can cooperate with cisplatin-induced tumor cell death. 3) Tier 2 supporting evidence: other T-cell exhaustion markers (CTLA4, LAG3, TIGIT) and additional matrisome genes (FN1, COL3A1, LAMA2) provide plausibility that immune dysfunction and stromal composition modulate the PLSP→response relationship; TSR and tumor budding may act as confounders/modifiers. 4) Why therapy_response endpoint: the project goal is to predict cisplatin-based neoadjuvant chemotherapy benefit in NMIBC (a treatment-decisions problem), so a therapy_response endpoint (pathologic downstaging/clinical responder vs non-responder) is the most relevant and actionable. 5) What I excluded and why: I excluded DNA-repair genes (e.g., ERCC family) and broad multi-gene predictive signatures from the core hypothesis to keep the mechanistic question focused on why immune spatial patterns sometimes fail; nuclear morphometry and TLS-specific constructs were deliberately left out of the core claim to avoid diluting the TIL–exhaustion–stroma mechanistic axis, though they can be tested as secondary features?

Working Hypothesis

In pretreatment NMIBC tumors treated with cisplatin-based NAC, high peritumoral TIL density predicts higher likelihood of pathologic response, but only when the local immune microenvironment is competent (low PDCD1 expression indicating reduced T-cell exhaustion) and the stromal barrier is modest (low COL1A1 expression). If PDCD1 is high (exhausted T cells) or COL1A1 is high (dense collagen barrier), the positive association between TIL density and response is attenuated or reversed. Therefore, the predictive value of TIL density for chemotherapy responsiveness is contingent on the functional state of infiltrating T cells and the degree of ECM-mediated barrier function, implying a composite biomarker that integrates immune state (PDCD1) and stroma (COL1A1) with TIL density. The model posits interaction effects: TIL_density × PDCD1 and TIL_density × COL1A1 significantly predict therapy response, and the three-way interaction TIL_density × PDCD1 × COL1A1 may refine risk stratification. Secondary modifiers (CTLA4, LAG3, TIGIT, FN1, COL3A1, LAMA2, ERBB2) will be explored as modulators of the core interaction. The endpoint is pathologic response to cisplatin-NAC, defined as downstaging to non-invasive disease or pCR, assessed on standardized post-treatment specimens. Measurement will be via digital pathology for TIL density, multiplex IHC and/or spatial transcriptomics for PDCD1 and COL1A1 in the peritumoral zone, with analyses adjusted for clinicopathologic covariates. Feasibility: use retrospective NMIBC cohorts with pre-treatment samples and known NAC outcomes; validate findings in an independent cohort; perform sensitivity analyses for threshold definitions and potential confounders.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	PDCD1, COL1A1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PDCD1: PDCD1 as an anchor for T-cell functional state) — because exhausted T cells may be non■functional despite high density, and (b) stromal extracellular matrix barrier (COL1A1 as an anchor for collagen-rich stroma) — because dense collagen can impede drug delivery and immune–tumor contact.

COL1A1: COL1A1 as an anchor for collagen-rich stroma) — because dense collagen can impede drug delivery and immune–tumor contact.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CTLA4	Supporting modifier from knowledge graph
Secondary	LAG3	Supporting modifier from knowledge graph
Secondary	TIGIT	Supporting modifier from knowledge graph
Secondary	FN1	Supporting modifier from knowledge graph
Secondary	COL3A1	Supporting modifier from knowledge graph
Secondary	LAMA2	Supporting modifier from knowledge graph
Secondary	ERBB2	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 2: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CTLA4: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- LAG3: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.